The Long(er) Term Effects of Promotions

Bohdan Leniuk*

John Munro Godfrey Sr. Department of Economics, University of Georgia

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Abstract

This paper studies how internal promotions shape wages over subsequent employer-to-employer (E2E) moves and unemployment-to-employer (U2E) transitions. Using panel data from the National Longitudinal Survey of Youth 1979 (NLSY79), I estimate event-study wage dynamics around promotions and track post-promotion wage changes across labor-market transitions. I find that promotions raise wages by about 17% on average and are followed by a sustained acceleration in wage growth. Promotion-induced wage gains largely survive E2E moves, but are selectively undone after unemployment: workers with a prior promotion experience an average 12% wage loss upon re-employment, while otherwise similar non-promoted workers exhibit no average wage loss. Moreover, wage losses for promoted workers increase sharply with unemployment duration (about 0.97 percentage-points per month), compared with a much flatter profile for non-promoted workers (about 0.43 percentage-points per month). To interpret these facts, I develop a model of promotions with asymmetric learning in which the incumbent firm observes worker performance, while the outside market infers ability from coarse, publicly observed career outcomes (promotion, retention, and unemployment duration). In simulations, the model reproduces the differential wage responses to U2E transitions while still matching the immediate impact of promotions on wages.

JEL Codes: D82, J24, J31, J62, J64, M51
Keywords: Promotions, firm learning, asymmetric information, wage dynamics, job mobility, unemployment duration

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1 Introduction

In this paper, I study how internal promotions affect workers' subsequent wages across employer-to-employer (E2E) and unemployment-to-employer (U2E) transitions. Using panel data from the National Longitudinal Survey of Youth 1979 (NLSY79), I estimate wage dynamics around promotions and compare wage changes for workers with and without prior promotions. I then develop a structural model of promotions and asymmetric firm-learning, where course labor market outcomes (e.g. promotions and unemployment spells) serve as the source of information to unmatched firms.

Promotions are among the most salient publicly observable signals of within-firm advancement, yet many workers' careers also include both employer changes and unemployment spells. Understanding how the labor market values a promotion when a worker changes jobs—especially when a worker changes via displacement—is important for interpreting the long-run returns to career progression and for disciplining models that combine internal labor markets with learning.

My empirical analysis uses the NLSY79's promotion histories together with detailed employer, occupation, and wage records to measure both the immediate wage effect of promotion and the persistence of promotion-related wage gains through subsequent labor-market transitions. I complement these estimates with a structural framework in which the incumbent firm observes worker performance and chooses whether to promote, retain, or separate, while the outside market only observes coarse career outcomes (promotion/retention and unemployment duration) and updates beliefs accordingly. Guided by the data, worker type affects both on-the-job productivity and job-finding prospects, so that prolonged unemployment generates additional adverse inference.

The data reveal four main findings. First, promotions raise wages by about 17% on average and are followed by a sustained acceleration in wage growth. Second, wage changes at promotion are highly heterogeneous: among first promotions, the 10^{th} percentile wage response is roughly a 12% decline, while the 90^{th} percentile exceeds 50%. Third, promotion-related wage gains largely survive E2E moves: promoted and non-promoted workers exhibit similar wage changes when switching employers without intervening unemployment. Fourth, promotion-related wage gains are selectively undone after unemployment: previously promoted workers lose about 12% upon re-employment on average, while non-promoted workers exhibit no average wage loss. Moreover, the wage loss for promoted workers increases sharply with unemployment duration (about 0.97 percentage points per month), compared with a much flatter profile for non-promoted workers (about 0.43 points per month). These patterns are difficult to reconcile with a pure depreciation view of unemployment and instead point to an inference mechanism in which promotions change how subsequent unemployment spells are interpreted.

1.1 Related Literature

This paper contributes to three connected strands of work in labor economics and macroeconomics: the economics of promotions and internal labor markets, models of worker signaling and firm learning, and the empirical and theoretical literature on job separations and displacement. My goal in this section is not to provide an exhaustive survey, but to situate the paper's facts and mechanisms within widely used canonical frameworks in these areas and to clarify what is new about connecting promotions to subsequent outside-market outcomes.

Promotions and internal labor markets. A large literature studies promotions as part of internal labor markets, emphasizing the joint determination of job assignment, wage growth, and career ladders within firms. Classic evidence from personnel records documents that wages typically rise discretely at promotion and continue to grow faster thereafter, consistent with promotions reflecting both changing job tasks and evolving information about worker ability (Baker et al., 1994a,b)(see also the broader interal-labor market perspective in Lazear and Rosen, 1981). Gibbons and Waldman (1999) provide an influential model in which wage dynamics and promotions arise from learning about ability and the assignment of workers to jobs with different sensitivity to ability; this class of models helps interpret the life-cycle decline in promotion rates and the acceleration of wage growth around promotion.

Relative to this literature, my empirical contribution is to document promotion effects in a representative longitudinal survey (NLSY79) and to push beyond the within-firm lens by examining how promotions interact with subsequent employer transitions and unemployment spells. This focus is closer in spirit to modern search-and-wage-setting environments where workers move across employers and where career progression may have value (or cost) in the outside market, not just inside the current firm. A related and complementary literature studies job ladders and employer-to-employer transitions in frictional labor markets (e.g. Burdett and Mortensen, 1998; Postel-Vinay and Robin, 2002; Moscarini and Postel-Vinay, 2012), highlighting that wage growth over the life cycle is tightly linked to job-to-job mobility. My evidence that promoted and non-promoted workers behave similarly after employer-to-employer moves, but differ sharply after unemployment, provides a new set of moments for models that combine internal advancement with external search.

Signaling, firm learning, and what a promotion reveals. The second strand concerns how markets learn about worker ability and how observable career events serve as signals. Canonical models of employer learning include Jovanovic (1979) and Farber and Gibbons (1996), and the empirical approach in Altonji and Pierret (2001) uses the changing returns to observable skill proxies over experience to identify learning. In these frameworks, wages and employment histories aggregate information that helps firms update beliefs about productivity.

My model adopts the firm-learning logic but emphasizes an informational friction that is natural for promotions: the current employer observes rich within-job performance, while the broader market may only observe coarse public signals, such as the worker's job title or the fact of having been promoted. This idea relates to classic signaling arguments (Spence, 1973), but with the signal generated endogenously through an internal decision (promotion/retention/separation) rather than through an upfront investment. The paper is also related to work on statistical discrimination and inference from employment histories, where market beliefs depend on past outcomes that are only imperfectly informative about ability (e.g. Waldman, 1984; Greenwald, 1986). A key implication in my setting is that promotions can raise wages contemporaneously and yet make workers more exposed to adverse inference following unemployment, because the promotion changes what the market thinks the worker is *supposed* to be able to do and because unemployment duration is itself informative.

Job separation, unemployment duration, and earnings losses. The third strand studies job loss, unemployment, and subsequent wage outcomes. A well-known empirical finding is that displaced workers experience persistent earnings losses following separation, with large heterogeneity across workers and separation types (Jacobson et al., 1993; Couch and Placzek, 2010). In frictional labor markets, separation and unemployment are also central to matching and wage dynamics (Moscarini and Postel-Vinay, 2012), and a large literature emphasizes that unemployment duration can be informative about worker quality when firms have limited information (Lockwood, 1991; Kroft et al., 2013).

My empirical results speak directly to this literature by showing that wage changes following unemployment differ systematically by workers' prior promotion history: non-promoted workers exhibit little average wage loss after unemployment, while promoted workers experience sizable losses that are increasing in unemployment duration. This pattern is difficult to reconcile with a pure human-capital depreciation view of unemployment, and instead supports an inference-based mechanism in which the outside market updates beliefs using both the worker's prior career signal (promotion) and the subsequent signal embedded in the length of non-employment. In this sense, the paper contributes new evidence on heterogeneity in displacement costs and provides a tractable mechanism—promotion as a publicly observed, endogenous, and ability-correlated signal—that can be embedded in search models with learning to match these facts.

Finally, the model's emphasis on employer decisions as information to the market connects to a broader literature where labor market histories summarize private information and affect subsequent wage setting and sorting. In my framework, the promotion decision plays a dual role: it reallocates the worker within the firm and simultaneously changes the informational environment faced by the worker in any future unemployment spell. This link between internal advancement and external-market inference is central to interpreting the longer-run effects of promotions documented in the data.

1.2 Data

The primary data-source is the National Longitudinal Survey of the Youth (NLSY). The NLSY is currently split into two single-cohort panels: the NLSY79 and the NLSY97. I utilize only the '79 survey because the later does not collect information on promotions.

The NLSY79 The 1979 survey of the National Longitudinal Survey of the Youths (NLSY79) is a longitudinal survey of individuals of age 15-21 on January 1st, 1979. Respondents are surveyed annually from 1979-1994, and bi-annually from 1996-present. Each survey consists of both a retroactive (weekly) employment diary and a survey of demographics, wealth, etc. For tractability (both in empirics and in the subsequent model), I aggregate to the monthly frequency when developing the full panel.

Pivotal to this analysis, from 1988-2016, the employment diary includes the respondents account of his/her promotion history. More specifically, each respondents is asked if "[he/she] experienced an increase in responsibility in her job". If yes, they following up by asking "does [he/she] consider that a promotion". Due to this, I find that the NLSY's definition of a promotion is related to that defined in Bayer and Kuhn (2023), where they claim that a promotion is an increase in autonomy and responsibility.

Occupation reports is of some concern in the labor literature. Kambourov and Manovskii (2013, 2008) both find evidence that reported occupation changes in the PSID, CPS, and related employment panels over-report occupation changes due to issues with measurement / data collection. I, too, find this the case in the NLSY. In order to gather occupation information in the NLSY, surveyors are directed to interpret an occupational code based on the respondents description of their day-to-day activities. Thus, it is common for an occupation in survey 1 to be different than survey 2 simply because the interpretation changed. Kambourov and Manovskii (2013) remedy this concern by imposing one occupation per employer spell. I relax this and exploit the timing of occupational changes with respect to surveys. That is, if a worker's occupation changed sometime between surveys, it means the respondent purposefully reported them as separate jobs. Through this method, I identify a handful of "within-firm" occupation changes, though they are not particularly common.

Sample Selection To construct my sample from the NLSY79, I first filter only male respondents, and only keep those with valid demographic information. Following this, I drop all observations with missing employer, occupation, wage, and hours worked and all individuals with missing demographic information. Since promotions are provided only between the years 1988-1990 and 1996-2016, I drop observations with missing promotional information if they are in those years, and then drop observations with other various missing information (such as test scores). Similar to Guvenen et al. (2020), I drop observations for each individual before they enter the labor market for the first time, and drop observations for each individual prior to them leaving school. Following this, I drop all observations from any individual with response rates lower than 85%[1]. Table 1.1 provides details on this selection process. The final panel consists of 2,193 individuals and 1,899,411 person-month observations.

Selection Criteria	**Individuals**	**Observations**
	12,686	6,698,208
Only Males	6,403	3,380,784
Missing Demographics	6,393	3,375,504
Missing Employment	6,314	2,234,942
Missing Promotions	6,314	2,234,238
Other Missing	5,260	1,899,411
Post-School	4,568	1,408,944
Post Labor Market Entry	3,199	1,232,946
Response Rate $\geq$ 85%	2,193	1,899,411

Notes: Demographics include race and education information. Missing employment means either occupation or employer codes are missing. Missing promotions are only dropped if between 1988-1990 and 1996-2016 since they are only reported for those years. Other missing information includes missing test scores, missing wages, or discrepancies between wages various responses.

Table 1.1: Sample Selection

2 Empirics

This section documents the incidence and wage effects of promotions and relates promotions to subsequent employer transitions, with and without intervening unemployment spells. Section 2.1 characterizes promotions and estimates their contemporaneous and distributional effects on wages. Section 2.2 uses the panel structure of the NLSY to study how promotions shape subsequent job mobility and outcomes. Section 2.3 summarizes the empirical findings and connects them to the theory in section 3.

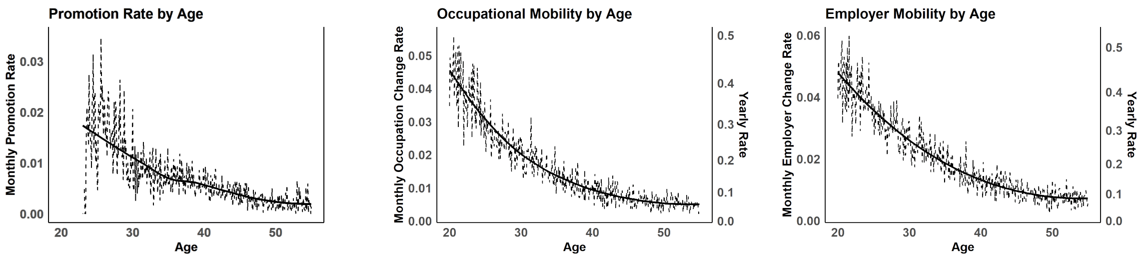
2.1 Characteristics of Promotions

Promotions are frequent early in the life cycle and decline sharply with age. Panel A of Figure 2.1 plots the monthly promotion rate by age. At the start of a career, the average worker is promoted with probability $\sim$ 2% per month (20% per year). By age 35, the rate falls to under 1% per month (10% per year) and approaches zero near the end of the career[2]. Panels B and C show analogous declines in occupational and employer churn, respectively.

[1] Changing the response-rate restriction has little effect on the observed empirics discussed in section 2.

[2] This trend is true across the education gradient, though more highly educated are positively shifted relative to those with less education

A natural interpretation of these profiles is learning: over time, workers and firms acquire information about match quality and comparative advantage, reducing the need for reassignment. Gervais et al. (2016) and Papageorgiou (2014) argue that this decline in occupational churn reflects workers becoming progressively better informed about which occupations suit their skills[3].



(a) Promotion (b) Occupation (c) Employer

Figure 2.1: Life-Cycle of Employment Mobility

Notes: Panel A-C provide the monthly rate of promotions, occupation changes, and employer changes for each age (by month). Dashed lines are raw data. Solid is local-linear smoothing. Along the RHS is a transformation of the monthly rate for interpretability

Unlike occupation and employer changes, promotions are absent for a sizable minority of workers. In the NLSY, only 4.17% and 2.73% of worker go through their entire career without changing employers or occupations, respectively, and the average worker changes occupations nearly 6 times and employers 7.6 times. In contrast, just over 30% of individuals never receive a promotion, and nearly 30% receive only one. Figure 2.2 reports the full distribution.

Turning to wages, promotions are associated with large contemporaneous gains. The average employee's first promotion raises current wages by 17.7%, about \$2.7/hour. Previous empirical studies of promotions (Baker et al., 1994a,b; Gibbons and Waldman, 1999) find that wage growth often accelerates after promotion relative to the pre-promotion trend. The NLSY sample shows the same pattern, though more modestly: in the 36 months leading into a worker's promotion, wages rise on average from \$11.85 to \$13.14 (\$0.43/year); in the 36 months following the promotion, wages rise from \$15.29 to \$17.65 (\$0.73/year). Figure 2.3 summarizes these dynamics: Panel A plots wage levels around the first promotion and Panel B reports percent changes relative to the last pre-promotion wage.

Despite the large average affect, there is a large degree of heterogeneity in the effect of promotions on wage. Table 2.1 reports selected quantiles of the wage change at promotion. The lower tail is particularly striking: nearly 20% of promotions have a zero or *negative* contemporaneous wage effect, and the 10^{th} percentile experiences a wage decline exceeding

[3] Guvenen et al. (2020) and Lise and Postel-Vinay (2020) describe similar environments. Guvenen specifically finds evidence that worker-churn increases as the distance between the vector of the workers skills and the employers requirement increases.

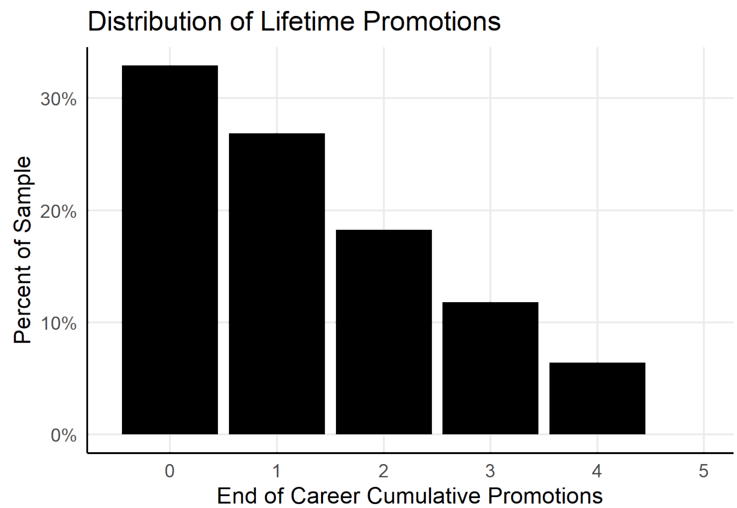


Figure 2.2: Distribution of Life-Time Promotions

Notes: Shows the % of the NLSY sample who recieve 0-5 promotions over the course of their career. Omitted are those who are promoted 6 or more times, of which there are relatively few.

Table 2.1: Distribution of Promotions

	p10	p20	p50	p80	p90
Wage Growth (%)	-12.2	1.57	14.73	34.81	51.76

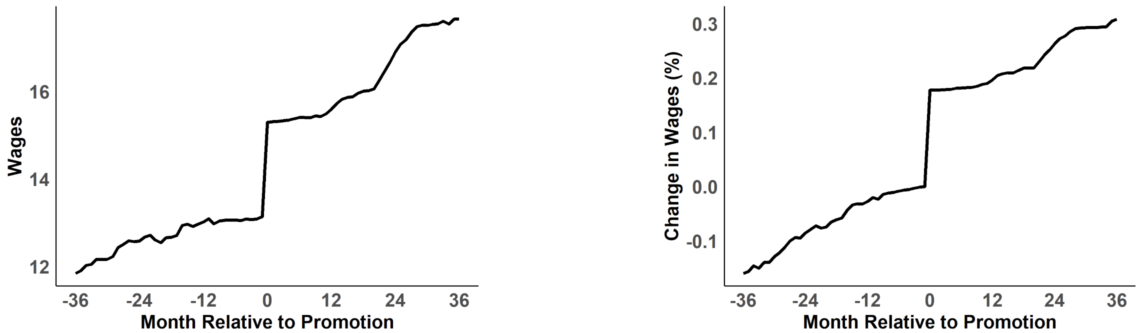
Notes: Values represent the 10^{th}, 20^{th}, 50^{th}, 80^{th}, and 90^{th} percentiles of promotion-induced wage gains.

10%. At the upper tail, the 90^{th} percentile gains about 50%.

Gibbons and Waldman (1999) argues that getting promoted is a job-assignment into a different production function. Firms promote workers as their human capital grows beyond the threshold where production in a promoted position exceeds that of production in the unprompted position. Assuming that wages track productivity in some way, dispersion such as in Table 2.1 suggests that motive other that production motivates both an employers incentive to offer a promotion and a workers incentive to accept. In the section 3, I assume that this is taste that takes the form of a T1EV error across positions.

Who gets promoted is central for interpreting these patterns. Observable skills matter, but the data point to substantial time-invariant heterogeneity in promotion propensity. I quantify this heterogeneity by estimating a fixed-effect logit model of promotions, given in equation 2.1.

$$\Pr(\text{Prom}_{it} = 1 \mid X_{it}, \alpha_i) = \Lambda\left(\mathbb{X}_{it}'\beta + \alpha_i\right), \qquad \alpha_i \sim \mathcal{N}(0, \sigma_\alpha^2). \tag{2.1}$$



(a) Hourly Wages

(b) Wage Changes (%)

Figure 2.3: Wage Effect of Promotions

Notes: Average wages plotted on an event horizon centered at a worker's first promotion. Panel A is hourly wages, while Panel B is log wages, normalized to event-time $t = -1$

Within $\mathbb{X}_{it}$ I include third order expressions of total worker experience, firm tenure, and occupational tenure. Additionally, it includes the lagged value of lifetime promotions at time t. The term α_i captures time-invariant individual heterogeneity in promotion propensity. I model this component as a normally distributed random intercept, $\alpha_i \sim N(0, \sigma_\alpha^2)$, which is assumed to be independent of the error and covariates in $\mathbb{X}_{it}$.

The purpose for the random effects model in the logistic setup is to exploit its ability to quantify explained heterogeneity from a rare (2% per month) 0-1 outcome through its estimation of σ_α^2. This analysis finds that over 92% of the variability in the propensity to be promoted is explained by the fixed effects alone.

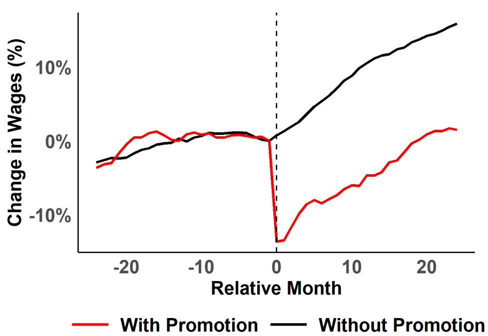
2.2 Promotions and Worker Churn

Turning to the relationship between promotions and broader labor-market dynamics, three facts stand out: (i) after losing a job, promoted individuals experience wage declines of $\sim 14\%$, whereas the effect of job loss for the non-promoted is close to zero; (ii) employer and occupation transitions that do *not* involve an intermittent unemployment spell affect the promoted and the non-promoted in a similar, positive way; and (iii) the effect of unemployment *duration* is nearly 0 for non-promoted individuals, but duration rapidly diminishes wages for those who were promoted. The remainder of this section documents and interprets these facts.

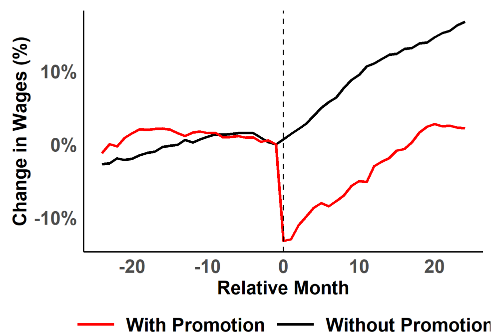
Throughout the remainder of this section, I refer to two groups of individuals: promoted and non-promoted. The promoted group consists of individuals who were promoted with their employer prior to the transition. For example, suppose an individual never received a promotion from his first employer, but was promoted with his second employer. Going from employer 1 to employer 2 would be considered a non-promoted transition, while going from employer 2 to employer 3 is a promoted transition.

Promotions and Job Loss I begin by studying how the consequences of job loss vary with prior promotion status. A large existing literature (Jacobson et al., 1993; Couch and Placzek, 2010; Davis and von Wachter, 2011; Schmieder et al., 2016)[4] studies separations from employment and their effects on worker outcomes. The consensus is that job loss has an immediate negative effect on wages, and losses grow as workers remain out of employment.

Separating the sample into promoted and non-promoted workers, the average wage decline through an unemployment spell is roughly 14% for the promoted and nearly 0% for the non-promoted. Figure 2.4 plots log wages, normalized to the last wage before an unemployment spell, over a 48-month window (24 months before job loss and 24 months after re-entry). Panel A includes all unemployment-to-employment (U2E) transitions, while panel B restricts to U2E transitions that also involve an occupation change[5].



(a) Employer Changes



(b) Occupation Changes

Figure 2.4: Wages Around U2E Transitions

Notes: Both panels plot average log-wages, normalized to a workers last pre-unemployment wage $t = -1$. Observations to the right of the dotted line are before an unemployment spell began, whereas observations to the right are after it began, thus for each worker, there is a varying amount of time between observation $t = -1$ and $t = 0$. The sample for panel A consist of all unemployment spells that satisfy (i) $\geq$ 24 months after a worker enters the panel, (ii) $\geq$ 24 months before he leaves the panel, and (iii) result in an employer change. Panel B's sample satisfy the same, and additionally must result in an occupation change.

Motivated by this descriptive evidence, I estimate equation 2.2 in an event-study design. Promoted workers are typically more tenured, more educated, and employed in more demanding occupations; failing to account for these differences would mechanically inflate the estimated role of promotion status.

[4]this is not an exhaustive list

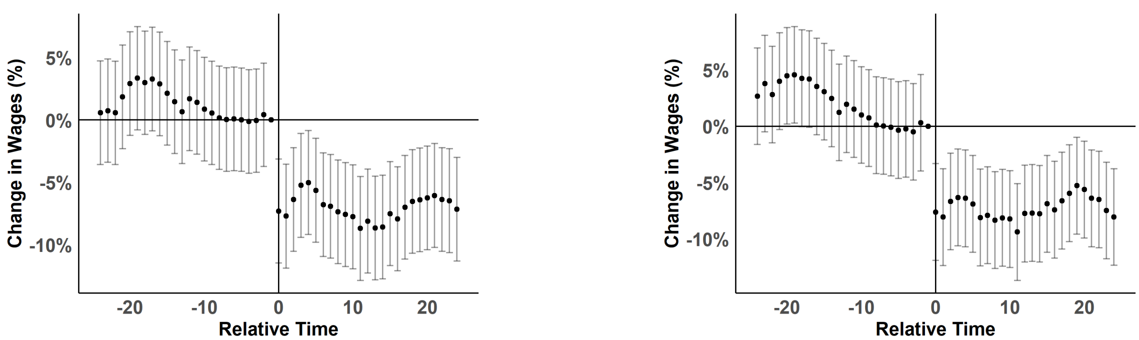
[5]Occasionally, workers lose their job and return to work with the same employer. These cases are not included in either panel

$$\Delta \log w_{its} = \sum_{\substack{k=-24 \\ k\neq -1}}^{24} \beta_k \mathbf{1}\{\tau_{its} = k\} \operatorname{Prom}_{is} + \sum_{\substack{k=-24 \\ k\neq -1}}^{24} \gamma_k \mathbf{1}\{\tau_{its} = k\} + \alpha_{is} + \lambda_t + \beta_{\mathbb{X}} \mathbb{X}_{its} + \varepsilon_{its} \quad (2.2)$$

$$\text{where, } \Delta \log w_{its} \equiv \log w_{its} - \log w_{i,t-1,s},$$

In it, i indexes workers, s indexes employment spells (with $s = 1$ denoting worker i's first post-unemployment spell), and t denotes months since re-entry into employment following spell s. Included in $\mathbb{X}_{its}$ is a third order polynomial in age, employer tenure, occupational tenure, continuous employment tenure, and experience. $\mathbb{X}_{its}$ also includes indicators for the 2-digit SOC occupation code and a linear term both in unemployment duration and occupational cognitive requirements as defined by Guvenen et al. (2020). Included in the α_{is} term is person $\times$ spell fixed effects[6], and λ_t is the time fixed effect.

The estimates of β_k are reported in Figure 2.5. As expected, controlling for loss of tenure and occupational "hierarchy" attenuates the gap, but the overall effect remains sizable: promoted individuals lose uniquely more wages ($\sim 7.5\%$ more) than do their non-promoted counterparts. As in Figure 2.4, I report results for all U2E employer changes (panel A), and for the sub-sample that also involves an occupation change (panel B). The results are nearly identical across the specifications.



(a) Employer Changes

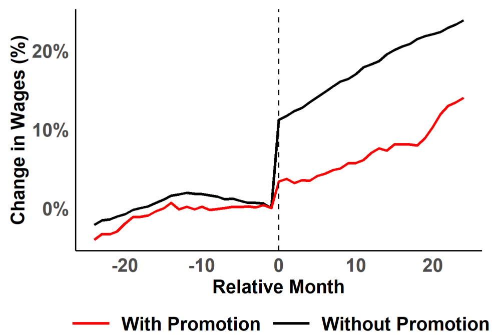
(b) Occupation Changes

Figure 2.5: U2E Event Study - Effect of Promotions

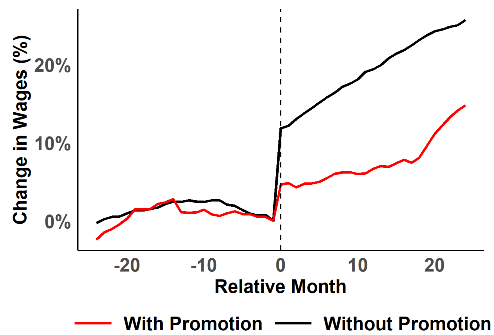
Notes: Plotted are the Promotion $\times$ relative-month fixed effects around an unemployment spell. Significant negative coefficients for $t \geq 0$ indicate that promoted individuals lose more wages in unemployment than the non-promoted. Regressions control for third-order polynomials in tenure, age, and experience; indicators for 2-digit SOC codes, and a linear term in both unemployment duration and occupation cognitive requirements (Guvenen et al., 2020). Panel A and B are separated as described in Figure 3.4

[6] Specifications with only individual fixed effects do not change the results

Promotions and E2E Transitions When transitions do not involve unemployment, differences by promotion status are less pronounced. On average, an employer-to-employer (E2E) transition is associated with nearly a 5% wage gain for promoted workers and around a 12% gain for non-promoted workers. The gap (about $\sim 7\%$) is roughly half the size of the gap associated with unemployment transitions (about $\sim 14\%$), and both groups experience positive wage gains. Figure 2.6 replicates the event-time plot in Figure 2.4, centered at an E2E transition.



(a) Employer Changes



(b) Occupation Changes

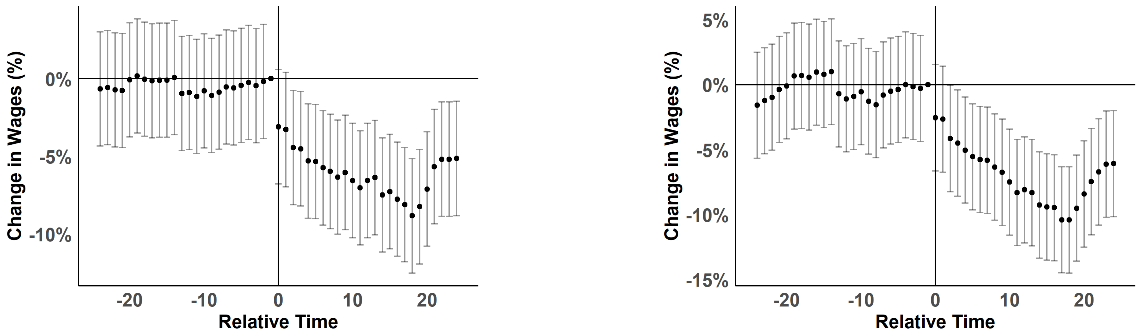
Figure 2.6: Wages Around E2E Transitions

Notes: Panels A and B both plot average log-wages, normalized to time $t = -1$, in a 24 month window centered around a workers employer-to-employer transition. Panel B plots a subset of workers in panel A, whose employer change also coincided with an occupational change.

I estimate a counterpart to equation 2.2, where s instead indexes a worker's s^{th} E2E transition. Figure 2.7 reports the resulting set of β_k. Conditional on observables and fixed effects, promoted and non-promoted workers exhibit broadly similar immediate wage dynamics around employer changes *without* intervening unemployment. Notably, however, wage growth for the non-promoted group outpaces that of promoted workers in the subsequent two years. Consistent with upward job ladders, I find that the non-promoted group's promotion rate rises over this horizon, suggesting that some E2E moves reflect search for positions with higher promotion prospects.

Promotions and Unemployment Duration Finally, I study heterogeneity in job-loss costs by unemployment *duration*. Unlike non-promoted individuals, the wage-loss function for promoted workers is strongly duration-dependent. Each additional month of unemployment reduces the next wage of a promoted worker by nearly 0.97 percentage points on average, compared with roughly 0.43 percentage points for the non-promoted. Figure 2.8 provides a visual summary of this pattern.

A promoted worker who enters unemployment and re-enters very quickly experiences an average loss of about 7.5% of his previous wage, while a similar non-promoted individual



(a) Employer Changes (b) Occupation Changes

Figure 2.7: E2E Event Study - Effect of Promotions

Notes: Plotted are the Promotion × relative-month fixed effects around an employer-to-employer (E2E) transition. Panels A and B are separated as described in Figure 2.4. The specification is identical to that in Figure 2.5, where the only difference is the center of the horizon.

sees no change. The gap grows by more than 2X by 18 months, the 80^{th} percentile of duration in the NSLY[7].

I estimate a flexible version of the wage-loss function in equation 2.3. Where i is an individual, s represents the first month of work following a worker's s^{th} unemployment spell, and $\Delta \log w_{is}$ is the difference in log wages immediately before and immediately after a worker's s^{th} unemployment spell. Included in $\mathbb{X}_{is}$ contains a workers experience, age, and tenure the immediately prior to the unemployment spell, and Req_Rank, the difference in occupational skill requirements (Guvenen et al., 2020) of his pre and post unemployment occupation.

$$\Delta \log w_{is} = \alpha + \beta_1 Prom_{is} + \beta_2 U_{is} + \beta_{\mathbb{X}} \mathbb{X}_{is} + \beta_3 Prom_{is} U_{is} \tag{2.3}$$

The results from this specification are provided in Table 2.2. The estimates mirror Figure 2.8: promoted individuals are uniquely harmed as an unemployment spell prolongs (around 0.7 p.p. per month), whereas non-promoted workers exhibit a much smaller negative effect (around 0.2 p.p. per month). Column (1) reports the baseline specification and column (2) restricts to unemployment spells that also involve an occupational change.

2.3 Discussion

In sum, several empirical facts stand out: (i) promotions generate large jumps in contemporaneous wages ($\sim 17\%$) and, once promoted, wage growth for an individual increases nearly 100%; (ii) promotion-induced wage changes are highly dispersed, with the 10^{th} percentile implying a wage *decrease* of $\sim 12\%$ and the 90^{th} percentile implying an increase

[7] Recall, duration is not limited to months the CPS considers unemployed. In fact, a majority of *non-employment* in the NLSY is not reported as unemployment

Table 2.2

	Dependent variable:	
	(1)	(2)
Prom	−0.011	−0.024
	(0.024)	(0.026)
UnemCount	−0.002***	−0.001**
	(0.001)	(0.001)
Exp	−0.013	−0.027
	(0.014)	(0.016)
Exp^2	0.0003	0.001
	(0.001)	(0.001)
Exp^3	0.00000	−0.00002
	(0.00002)	(0.00002)
Age	0.015	0.062
	(0.052)	(0.060)
Age^2	0.0002	−0.001
	(0.001)	(0.002)
Age^3	−0.00000	0.00001
	(0.00001)	(0.00001)
Tenure	−0.029***	−0.021**
	(0.007)	(0.009)
Tenure^2	0.001	0.001
	(0.001)	(0.001)
Tenure^3	−0.00002	−0.00001
	(0.00002)	(0.00003)
Req_Rank	0.089***	0.095***
	(0.028)	(0.029)
Prom:UnemCount	−0.005***	−0.004***
	(0.001)	(0.001)
Constant	−0.309	−0.853
	(0.629)	(0.719)
Observations	5,406	4,147
R^2	0.046	0.053
Adjusted R^2	0.040	0.045
Residual Std. Error	255.882 (df = 5371)	261.593 (df = 4112)
F Statistic	7.698*** (df = 34; 5371)	6.757*** (df = 34; 4112)

*p<0.1; **p<0.05; ***p<0.01

Notes: (1) regresses the percent change in wage for individuals who moved into unemployment. (2) does the same, but restricts the sample to those that also changed occupations. Req_Rank is the difference in percentile of the post-unemployment occupation from the pre-unemployment occupation.

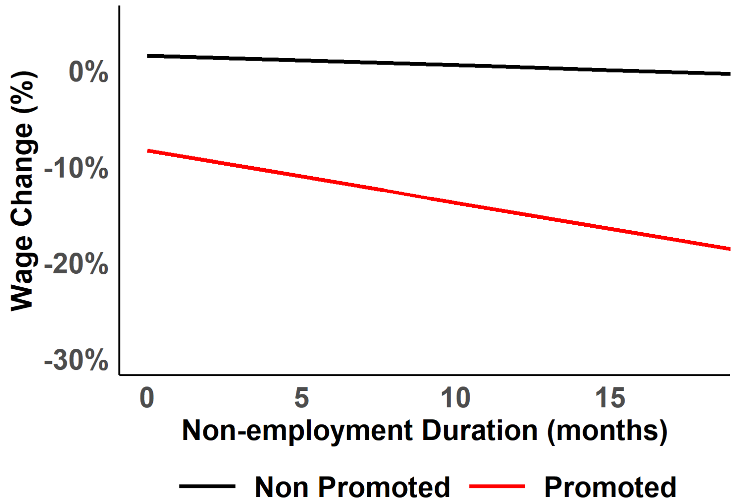


Figure 2.8: Wage Loss as a Function of Duration and Promotion

Notes: Plots an estimated linear-trend of wage loss and as a function of promotion status (before job loss) and duration of the unemployment spell.

of $\sim 55\%$; (iii) observable characteristics explain relatively little of promotion propensity, while individual fixed effects account for much of the observed variance; (iv) promoted individuals experience much larger wage declines in unemployment than the non-promoted, but both groups are similarly affected by E2E transitions; and (v) for promoted individuals, unemployment-induced wage loss is highly dependent on unemployment duration, whereas duration matters relatively little for the non-promoted.

I argue that these facts can be explained by a model of promotions and firm learning. Firms assign workers to positions based on beliefs about imperfectly observable skill and update these beliefs using workers' histories of promotions and job separations. Section 3 introduces the model, and section 4 discusses estimation and model fit.

3 Model

This section lays out a model of promotions and their interaction with the broader labor market. There are two types of agents—workers and employers. Section 3.1 describes the environment and worker side of the model, section 3.2 describes employers and their beliefs, and section 3.3 defines the equilibrium. Time is discrete.

3.1 Workers

There is a continuum of heterogeneous workers. A worker's type $z \in \{0, 1\}$ is fixed over the life cycle, where $z = 1$ (high-type) workers are more productive both in work and in job search. Type is privately known to the worker and imperfectly observed by employers, thus, employers form expectations about a worker's type. Upon matching with an employer, a worker draws match quality $q \in \{0, 1\}$ with probability λ. I assume that z and q are perfect substitutes in production.

Beliefs are denoted by π and χ, where $\pi = P_j(\tilde{z}_i = 1)$ and $\chi = P_{-j}(z_i = 1)$. Here i indexes the worker, j is the worker's *current* employer, and $-j$ denotes all other (non-matched) employers. A matched employer tracks $\tilde{z} = \max\{z, q\}$, whereas non-matched employers track only z. This need not be the case, but it simplifies both notation and computation and comes at no theoretical expense.

Each period, workers exit the model with probability d and are replaced by new entrants whose type is drawn from the unconditional distribution Π_z.

Employment Workers are either employed ($e = 1$) or unemployed ($e = 0$). Conditional on employment, the matched employer assigns a promotion status $p \in \{0, 1\}$, where $p = 0$ denotes a non-promoted position and $p = 1$ a promoted position. While employed, workers are paid a wage $w(\pi, p) = \omega \mathbb{E}[y(\tilde{z}, p) - k_p | \pi]$, where $y(\tilde{z}, p)$ is production, $\omega \in [0, 1]$ is the wage rate, and k_p is a fixed cost of maintaining the match[8]. Production is given by equation 3.1. Here, $\varepsilon \sim N(0, \sigma_y^2)$ is white noise and c_p is the penalty from assigning a worker to an ill-suited promotion status.

$$\log(y) = \varepsilon + \tilde{z} - c_p |p - \tilde{z}| \tag{3.1}$$

Equation 3.1 implies complementarity between promotions and high types/quality matches: misassignment reduces output through $c_p|p - \tilde{z}|$. Low-type workers are less productive in promoted positions, and high-type workers are less productive when not promoted. Empirically, Guvenen et al. (2020) and (Lise and Postel-Vinay, 2020) document sizeable production losses from mismatch between worker skills and job assignments. The intuition is that low-type workers do not have the necessary skills, and that high-type workers are not maximizing their potential without the added responsibilities.

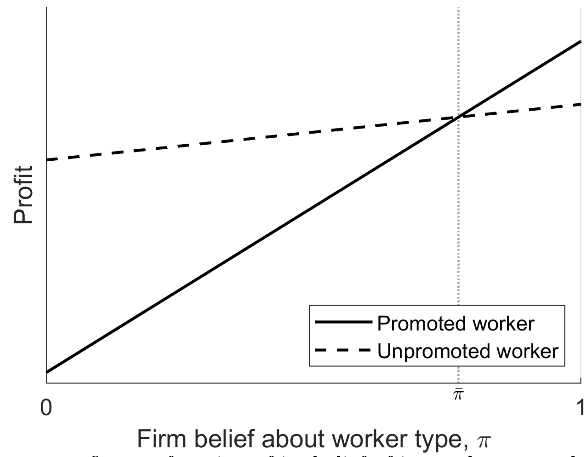
The specification is consistent with standard promotion models. In Gibbons and Waldman (1999), production is linear in general human capital, and the assumptions $y_0(0) > y_1(0)$ and $y_1'(h) > y_0'(h) \ \forall \ h$ deliver a single-crossing condition: the employer promotes when human capital (h) exceeds a threshold $\bar{h}$ satisfying $\bar{h} := y_0(\bar{h}) = y_1(\bar{h})$. Under equation 3.1 and the wage rule, the employer's promotion decision similarly exhibits single crossing in π, as illustrated in Figure 3.1.

At the end of each period, an employed worker separates exogenously into unemployment with probability δ. Conditional on no separation, the worker is reviewed with probability ρ. A review triggers Bayesian updating of the matched employer's belief π and a subsequent decision to fire, promote, or retain the worker in the current position. I rule

[8]A fine interpretation of k_p can be rent or opportunity cost

out demotions, both because they are rare in the data ($< 1\%$ per year) and because the restriction substantially simplifies computation.

Figure 3.1: Employer Profit as Function of π, p



Notes: Employers profit as a function of its belief of its worker, π, and the promotion status it assigns to him. The intersection, $\bar{\pi}$, is not guaranteed to be in $[0, 1]$, but in estimation, $\bar{\pi}$ will be used to target the total promotion rate.

The value functions for an employed non-promoted worker and an employed promoted worker are provided in equations 3.2 and 3.3, respectively. In them, each of Γ_P, Γ_{NP}, and π' are belief-transition functions, and are defined in section 3.2. The value for an unemployed worker is given by V_u.

$$V_e^{NP}(z, q, \pi, \chi, nu) = w(\pi, 0) + \beta(1-d)\begin{pmatrix} \delta V_u(z, \Gamma_{NP}(nu+1, \chi))+ \\ (1-\delta)\rho\mathbb{E}\Big[F(\pi'(y \mid \pi, p=0), 0)V_u(z, \Gamma_{NP}(nu+1, \chi))\Big]+ \\ (1-\delta)(1-\rho)V_e^{NP}(z, q, \pi, \chi, nu+1)+ \\ (1-\delta)\rho\mathbb{E}\Big[(1-F(\pi'(y \mid \pi, p=0), 0))(1-Pr(\pi'(y \mid \pi, p=0)))V_e^{NP}(z, q, \pi'(y \mid \pi, p=0), \chi, nu+1)\Big]+ \\ (1-\delta)\rho\mathbb{E}\Big[(1-F(\pi'(y \mid \pi, p=0), 0))Pr(\pi'(y \mid \pi, p=0))V_e^{P}(z, q, \pi'(y \mid \pi, p=0), \chi, nu+1, 0)\Big] \end{pmatrix} \tag{3.2}$$

$$V_e^{P}(z, q, \pi, \chi, nu, np) = w(\pi, 1) + \beta(1-d)\begin{pmatrix} \delta V_u(z, \Gamma_P(nu, np+1, \chi))+ \\ (1-\delta)\rho\mathbb{E}\Big[F(\pi'(y \mid \pi, p=1), 1)V_u(z, \Gamma_P(nu, np+1, \chi))\Big]+ \\ (1-\delta)(1-\rho)V_e^{P}(z, q, \pi, \chi, nu, np+1)+ \\ (1-\delta)\rho\mathbb{E}\Big[(1-F(\pi'(y \mid \pi, p=1), 1))V_e^{P}(z, q, \pi'(y \mid \pi, p=1), \chi, nu, np+1)\Big] \end{pmatrix} \tag{3.3}$$

Unemployment While unemployed, workers consume home production b and choose search effort $\theta \in [0, \infty]$ to maximize the present value of expected utility. Search effort maps into a job-finding probability via $p(\theta, z) = 1 - e^{-\alpha_z \theta}$, where $\alpha_1 > \alpha_0$, so high types convert effort into offers more effectively. Effort entails utility cost $\nu(\theta) = \xi\theta^{\gamma}$, where $\xi > 0, \gamma > 1$.

Equation 3.4 gives the unemployed value function for a type-z worker facing market belief χ. The objects V_P and V_{NP} denote the values upon entering employment in a promoted and non-promoted position, respectively, conditional on type z, internal belief π, and external belief χ.

$$V_u(\chi, z) = \max_\theta b - \nu(\theta) + \beta(1-d) \begin{pmatrix} p(\theta, z)Pr(m(\chi))\mathbb{E}_q\Big[V_e^P(z, q, \lambda + (1-\lambda)m(\chi), m(\chi), 0, 0)\Big] + \\ p(\theta, z)(1 - Pr(m(\chi)))\mathbb{E}_q\Big[V_e^{NP}(z, q, \lambda + (1-\lambda)m(\chi), m(\chi), 0)\Big] + \\ (1 - p(\theta, z))V_u(n(\chi), z) \end{pmatrix} \tag{3.4}$$

In equation 3.4, $m(\chi)$ and $n(\chi)$ map the prior market belief χ into the post-search belief after success and failure, respectively (defined in section 3.3). The function $Pr(\pi)$ maps the matched employer's belief into the promotion probability, allowing "promotion" upon hiring[9]. Note that due to q, if the worker matches with a firm, the initial π is scaled up from $m(\chi)$ by to reflect the possibility of a good match.

3.2 Employers and Beliefs

There are infinitely many homogeneous employers. When not matched with a worker, an employer posts vacancies at no cost, which are filled randomly. Therefore, the external market can be summarized by the set of $m(\chi), n(\chi) : [0,1] \to [0,1]$, $\Gamma_{NP}(\chi, nu)) : [0,1] \times \mathbb{N} \to [0,1]$, and $\Gamma_P(\chi, nu, np)) : [0,1] \times \mathbb{N}_0 \times \mathbb{N} \to [0,1]$, which are functions that map initial beliefs and employment outcomes to new beliefs. I first begin by describing matched employers and their belief process, and then move into a discussion on the external belief process.

Matched Employers At the end of each period, with a random probability ρ, a matched employer reviews his employee's performance. A performance review at the end of period t means to observe production for that period only, y_t. Upon this review, the employer updates his beliefs according to Bayes Rule. Equation 3.5 defines the update process, where the updated belief, π', is a function of the previous belief π and the promotion status of the worker. $f(y|z,p)$ is the density function production conditional on a worker's type and his promotion status.

$$\pi' = \pi'(y|\pi, p) = \frac{\pi f(y|z=1, p)}{\pi f(y|z=1, p) + (1-\pi)f(y|z=0, p)} \tag{3.5}$$

[9]though a promotion of this form would not be observed in the data, and will not be counted as a promotion in simulation

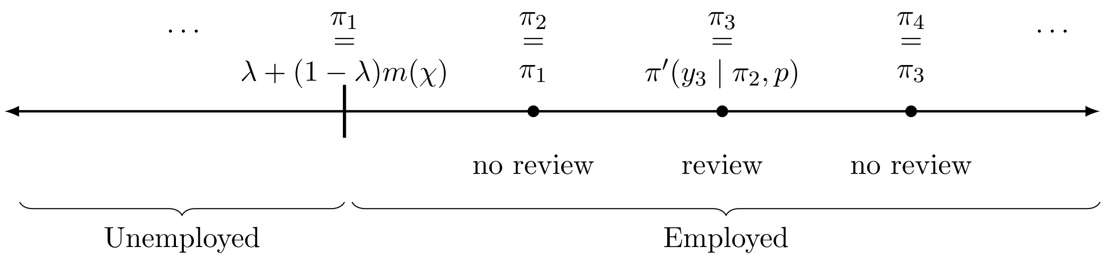


Figure 3.2: Evolution of π

Notes: This figure provides an example for how π, his current employer's belief of him, evolves over time. π' is a function defined in Equation 3.5.

Following the review, the employer makes two decisions: (i) whether to fire the worker (yielding value 0) and (ii) conditional on retention, whether to promote the worker if he is not already promoted. The firing and promotion decisions are given in equations 3.6 and 3.7, respectively.

$$F(\pi, p) = \begin{cases} 1 & \text{if } (1-\omega)\mathbb{E}[y(z,p) - k_p | \pi, p] < 0 \\ 0 & \text{otherwise} \end{cases} \tag{3.6}$$

$$P^*(\pi, p) = \max_{j \in \{p,1\}} \left\{ \beta^P((1-\omega)\mathbb{E}[y(z,j) - k_j | \pi, j] - p\psi) + \epsilon \right\} \text{ where } \epsilon \sim \text{T1EV} \tag{3.7}$$

In equation 3.7, ψ represents some training cost, incurred by the employer, for promoting its employee. Note that the choice correspondence for the decision to promote a worker is the set $\{p, 1\}$ which reflects the assumption that the firm cannot chose to demote a worker. Because $\epsilon \sim$ T1EV, the promotion rule for an employer is the equation 3.8.

$$\begin{aligned} \Pr(\pi) &= \frac{e^{\beta^P U(\pi)}}{1 + e^{\beta^P U(\pi)}}, \\ \text{where} \quad U(\pi) &= (1-\omega)(\mathbb{E}[y(z,1) - k_1 \mid \pi] - \psi - \mathbb{E}[y(z,0) - k_0 \mid \pi]) \,. \end{aligned} \tag{3.8}$$

Figure 3.2 provides an example of the evolution of π over time for a given worker. Upon the formation of the match, *pi* takes its initial value from the external labor market, scaled appropriately to account for the possibility that the worker is a matched well ($q = 1$). If a performance review does not occur, π simply persists into the next period. If a review does occur, *pi* evolves according to the equation π', taking production that period as the new source of information.

Non-Matched Employers The external labor market does not observe production directly, but it observes two outcomes that are informative about type: (i) job-finding success (or lack thereof) while unemployed and (ii) promotion histories across employment spells. Let $m(\chi)$, $n(\chi)$, $\Gamma_{NP}(\chi, nu)$, and $\Gamma_P(\chi, nu, np)$ map the market prior χ and observed histories into posterior beliefs. I begin by defining $m(\chi)$ and $n(\chi)$, which update beliefs following search success and failure.

Let $Pr(\pi)$ be the equilibrium promotion rule from equation 3.8 and let $h(z, \chi)$ be the equilibrium search policy from equation 3.4. Equations 3.9 and 3.10 define the corresponding belief updates: $m(\chi)$ is the posterior upon successful search, and $n(\chi)$ is the posterior upon unsuccessful search.

$$m(\chi) = \frac{\chi h(1,\chi)}{(1-\chi)h(0,\chi) + \chi h(1,\chi)} \tag{3.9}$$

$$n(\chi) = \frac{\chi(1-h(1,\chi))}{(1-\chi)(1-h(0,\chi)) + \chi(1-h(1,\chi))} \tag{3.10}$$

Updating χ from within-spell outcomes (e.g., a promotion) is not recursive because later events depend on intermediate performance signals. For example, consider a worker hired in period t who is promoted at the end of period $t+2$, and assume $\rho = 1$. Let χ denote the market belief at the start of the spell. Equation 3.11 gives the likelihood of this event.

$$\mathcal{L}(\chi, z) = \int_{-\infty}^{\infty} (1 - Pr(\pi'(y_1|\pi = \chi, 0)))(1 - F(\pi'(y_1|\pi = \chi, 0), 0)) \int_{-\infty}^{\infty} \cdots \\ Pr(\pi'(y_2|\pi = \pi'(y_1|\pi = \chi, 0), 0))\phi(y_2)dy_2 \;\; \phi(y_1)dy_1 \tag{3.11}$$

Because the period-$t+1$ outcome is nested in the promotion probability at $t+2$, the external market cannot update recursively within the spell. Instead, it treats the entire spell as a single event and computes $\mathcal{L}$ when the worker enters unemployment[10].

Under the no-demotion assumption, the sufficient state variables for this update are χ, nu, and np. Here nu is the number of periods spent non-promoted and np is the number of periods spent promoted. Without this restriction, the external market would need the entire promotion-status history within the spell, a vector of size 2^K where K is tenure in periods.

Under the no-demotion restriction, both $\Gamma_{NP}(\chi, nu)$ and $\Gamma_P(\chi, nu, np)$ can be defined as in equation 3.12 and 3.13, respectively.

[10]This is as if the external market reads the workers resume that contains job titles and dates

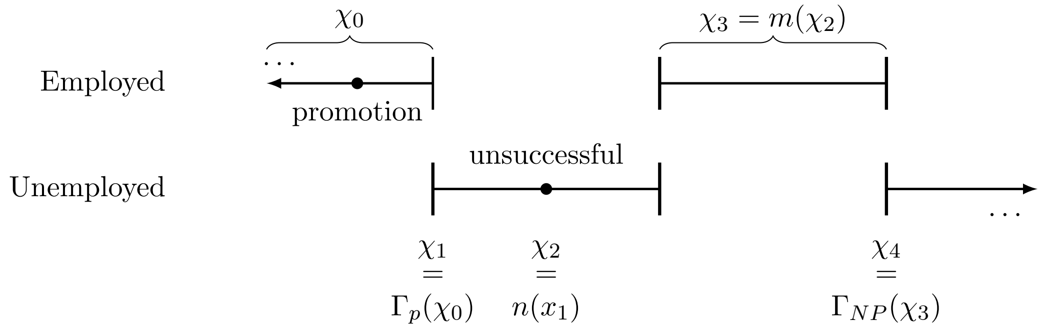


Figure 3.3: Evolution of χ

Notes: This figure provides an example of how χ, the "external market" belief of a worker, evolves over his career.

$$\Gamma_{NP}(\chi, nu) = \frac{\chi P_N^{(1)}(nu|z=1,\tilde{\chi})}{\tilde{\chi} P_N^{(1)}(nu|z=1,\tilde{\chi}) + (1-\tilde{\chi}) P_N^{(1)}(nu|z=0,\tilde{\chi})} \tag{3.12}$$

$$\Gamma_P(\chi, nu, np) = \frac{\chi P_P^{(1)}(nu,np|z=1,\tilde{\chi})}{\tilde{\chi} P_P^{(1)}(nu,np|z=1,\tilde{\chi}) + (1-\tilde{\chi}) P_P^{(1)}(nu,np|z=0,\tilde{\chi})} \tag{3.13}$$

where $\tilde{\chi} = \lambda + (1-\lambda)\chi$ is the matched employer's initial belief, which accounts for the probability λ that the match itself is high quality ($q = 1$).

$P_N^{(\tau)}(nu|\chi, z)$ is the probability that a type-z worker, matched with initial belief χ, is never promoted, works for nu periods, and then separates into unemployment. Similarly, $P_P^{(\tau)}(nu, np|\chi, z)$ is the probability that the worker spends nu periods non-promoted and then np periods promoted before separation. Formal definitions are provided in appendix A. Except for the inclusion of λ, this is the standard Bayesian update where the entire employment spell is treated as a single event.

Figure 3.3 provides an example of how χ evolves over time for a given worker. During employment spells, χ remains unchanged (χ_0 and χ_3 in the provided example). Upon job separation, χ updates according to the proper Γ function. For example, if he was promoted, as in the first employment spell in Figure 3.3, it evolves according to Γ_P. While unemployed, χ evolves after each period. Unsuccessful job searches evolve according to n, while successful search results in an update of χ according to m.

3.3 Equilibrium

Equilibrium is defined as a set of search policy functions $h(z, \chi)$, a promotion rule $\Pr(\pi)$, a firing rule $F(\pi, p)$, employed and promoted value function $V_e^P(z, \pi, \chi, nu, np)$, employed and non-promoted value function $V_e^{NP}(z, \pi, \chi, np)$, unemployed value function $V_u(z, \chi)$, and belief transition functions $n(\chi)$, $m(\chi)$, $\Gamma_P(nu, np, \chi)$, and $\Gamma_{NP}(\nu, \chi)$, such that, given a production function $y(z, p)$ and a wage rule $w(\pi, p)$,

1. $h(z, \chi)$ solves the worker search problem in equation 3.4 $\forall z, \chi$, and has a resulting value functions $V_u(z, \chi)$, $V_e^{NP}(z, \pi, \chi, np)$, and $V_e^P(z, \pi, \chi, nu, np)$
2. $Pr(\pi)$ is the resulting choice probability function from equation 3.7
3. $F(\pi, p)$ is the solution to the firing rule in equation 3.6 $\forall \pi, p$
4. Given $F(\pi, p)$, $Pr(\pi)$, and $h(z, \chi)$, $n(\chi)$, $m(\chi)$, $\Gamma_P(nu, np, \chi)$, and $\Gamma_{NP}(nu, \chi)$ are the Bayesian belief updating functions for employers.

Solving this model is done via value function iteration. First, use $y(z, p)$ and $w(\pi, p)$ to solve $Pr(\pi)$ and $F(\pi, p)$. Using these, solve $\Gamma_{NP}(nu, \chi)$ and $\Gamma_P(nu, np, \chi)$ as written in appendix A. Guess a value for V_e^{NP}, V_e^P, V_u, and $h(\chi, z)$. Using the guess for $h(\chi, z)$, solve $n(\chi)$ and $m(\chi)$ as written in equations 3.9 and 3.10. Appendix B provides a proof of existence.

4 Estimation and Model Fit

The model is estimated in two stages. First, I calibrate a set of parameters to match those provided in prior literature or to match basic descriptive results. After this, I use method of simulated moments to estimate parameters concerning the promotion rules, job search, and wages.

Tables 4.1 and 4.2 report the calibrated and estimated parameters, respectively. This section briefly motivates the calibration choices and then provides a more detailed discussion of how the estimated parameters are identified.

Calibration The unconditional distribution of worker types, Π, is estimated empirically using k-means clustering. For each worker in the NLSY, I construct four summary statistics: (i) average wage, (ii) total lifetime earnings, (iii) the number of unemployment spells, and (iv) the average duration of unemployment spells[11]. With $k = 2$, the algorithm partitions workers into two mutually exclusive groups by minimizing within-cluster Euclidean distance in this four-dimensional space. I interpret the resulting clusters as "low type" and "high type" workers, and set Π equal to the share of workers assigned to the high-type cluster.

The worker's share of expected profit, ω, is set to 0.5 to loosely match Nash bargaining with equal bargaining power. The standard deviation of transitory shocks to log production, σ_y, is set to 0.5, in line with values commonly used in the literature.

[11] For the small number of workers who never enter unemployment, I set the average duration to zero.

The flow cost of maintaining a non-promoted position, k_0, is set to 0.355 so that firms are roughly indifferent between hiring a worker with an extremely low promotion posterior ($\pi \approx 0$) and remaining unmatched[12]. The parameters d and δ are chosen to match two moments in the NLSY: an average career length of 480 months (40 years) and an average of five unemployment spells over the life cycle. Home production (or unemployment benefits), b, is set to 40% of average wages. The probability of a performance review is set to $\rho = 1/12$, implying one review per year on average. Finally, γ is set to 1.5, and α_0 is normalized to 1.

The discount parameter, β, is low. A monthly β of 0.9 implies a year discount factor of nearly 0.40. Because there are no accumulated variables (e.g. capital), the effect of β plays only a mechanical role to guarantee convergence. Increasing β to 0.99 ($\sim$ 0.89 yearly) only affects the average duration of unemployment, and can be completely offset by increasing ξ in the estimation procedure.

Table 4.1: Calibrated Parameters

Parameter	Value	Description	Calibration Target
Π	0.38	% of high types	Estimated outside model
ω	0.5	wage piece rate	Mortensen and Pissarides (1994); Shimer (2005)
k_0	0.35	flow cost of un-promoted position (e.g. rent)	value of low-type worker to firm
σ_y	0.5	s.d. of output shocks	Abowd and Card (1989); Meghir and Pistaferri (2004)
γ	1.5	curvature of search cost	
ρ	$\frac{1}{12}$	Review probability	1 review a year
d	$\frac{1}{480}$	Prob. of death/retiring	avg. 40-year career
β	0.9	Discount factor	
δ	$\frac{5}{430}$	Exogenous separation probability	Average 5 separations per career
b	0.2	Unemployment benefits / home production	40% of average wages
α_0	1	Search-productivity of low-types	Normalized to 1

Notes: Table includes both calibrated parameters and parameters estimated empirically outside the model. Calibration targets left blank imply that the parameters are set arbitrarily, but qualitative and quantitative results are robust to their effect.

Estimation After calibration, I estimate the remaining parameters using the method of simulated moments (MSM). In total, I estimate eight parameters: (i) the one-time cost

[12] Equivalently, k_0 is normalized to make the value of a low-type hire near the outside option of a vacancy.

Table 4.2: Estimated Parameters

Parameter	Estimate	Description	Identifying Moments
ψ	0.13	fix-cost to promote	average promotion size
c_0	0.60	production penalty for "wrong" non-promotion	slope of wages pre-promotion
c_1	0.66	production penalty for "wrong" promotion	slope of wages post-promotion
β^p	10.80	Promotion-preference shocks	sd of promotions
k_1	0.65	flow-cost of promoted position	% promoted in first job
α_1	1.30	search productivity of high types	difference in av unemployment spell by type
ξ	0.37	utility cost of search	Average unemployment spell
λ	0.145	probability of good match	average U2E wage loss for promoted and non-promoted

Notes: Estimated parameters with their value, description, and their empirical target. All values are estimated via method of simulated moments.

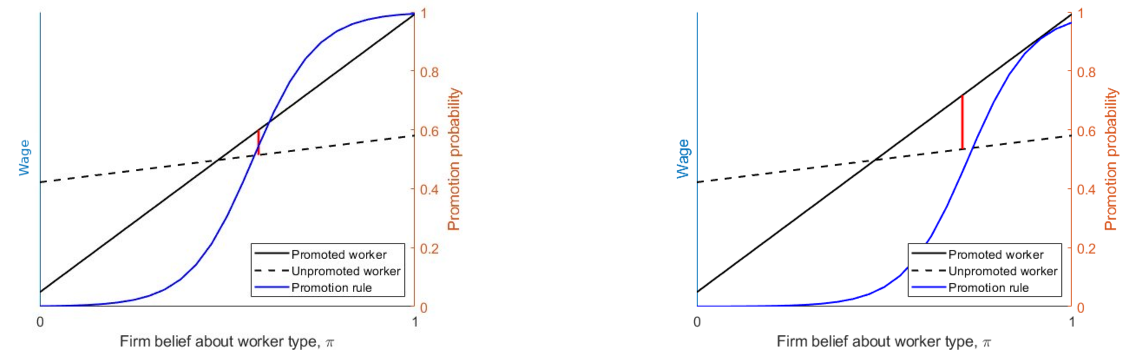
of promotion (e.g., training), ψ; (ii) the loss in log production from failing to promote a high-type worker, c_0; (iii) the loss in log production from promoting a low-type worker, c_1; (iv) the weight the firm places on profits in the promotion decision, β^p; (v) the flow cost of maintaining a promoted position, k_1; (vi) the search productivity of high types, α_1; (vii) the utility cost of search, ξ; and (viii) the probability that a new match is a "good fit", λ.

First, the set of parameters (ψ, c_0, c_1) are estimated to match Figure 2.3(b). The fixed-cost ψ primarily governs the average wage jump at promotion. Production penalties directly govern the slope of the wage function before and after a promotion. Figure 4.1 provides a visual representation of the identification of ψ. Given a wage function, ψ pushes the inflection point of the promotion function to the right. In both panels, the red line serves as a proxy for the median promotion size.

The parameter β^p is identified from the dispersion of wage changes at promotion. As shown in Table 2.1, the 10^{th} and 90^{th} percentiles of wage changes at promotion are -12.2% and 51.76%, respectively. In the model, β^p governs how sharply the firms promotion rule responds to changes in their belief of the worker. If $\beta^p \to \infty$, the promotion decision becomes deterministic: the firm promotes the worker as soon as promotion yields positive returns of profit, and never otherwise. If $\beta^p = 0$, the promotion functions is $0.5\ \forall \pi$, which generates excessive dispersion. Visually, decreasing β^P "flattens" the promotion function presented in Figure 4.1, which will clearly increase dispersion.

Given k_0, the parameter k_1 governs the value of $\bar{\pi}$ as discussed in Figure 3.1. Therefore, as k_1 increases relative to k_0, it becomes more rare for a worker to be promoted with any given employer.

The search parameters α_1 and ξ are estimated to match unemployment duration by



(a) "Small" Promotion Cost (b) "Large" Promotion Cost

Figure 4.1: Identification of ψ

Notes: Panel A and B provide the wage functions for a worker of belief π and promotion status. In blue represents the promotion function given the wage rule and the promotion cost, ψ. Average promotion roughly tracks where the promotion rule crosses the 50% mark. Panel A and B are identical other than the value for ψ, which is larger in B.

type. For this, I use the empirical type classification obtained from the aforementioned k-means procedure. After normalizing $\alpha_0 = 1$, α_1 governs the relative job-finding efficiency of high types and is therefore disciplined by the gap in average duration across types. The search-cost parameter χ shifts equilibrium search effort for all workers and is therefore pinned down by the overall level of unemployment duration.

Finally, λ is estimated to match the average wage change upon unemployment-to-employment (U2E) transitions (Figure 2.4). In the model, λ scales initial priors upon a match according to

$$\pi = \lambda + (1 - \lambda)m(\chi)$$

which mitigates the negative effect of unemployment specifically for workers with low initial χ. Conversely, λ scales *down* the value of Γ_P as shown in equation 3.13, which exacerbates the negative effect of unemployment for worker who were promoted.

Model Fit The model matches the wage effects of promotions reasonably well. Figure 4.2 plots log wages in a 36-month window around each worker's first promotion. The solid lines reproduce the empirical event study from Figure 2.3(b), while the dashed lines show the model simulation. The model slightly overestimates the average effect of promotion (just below 20% in the simulation versus 17.75% in the data), but it matches the average wage-growth rates before and after promotion reasonably well. However, it fails to fully match the wage path over the entire 36-month post-promotion window, which is not directly targeted.

The parameter β^p targets the dispersion of promotion-induced wage changes in the NLSY. Specifically, β^p is set to match a standard deviation of 0.3496. The model system-

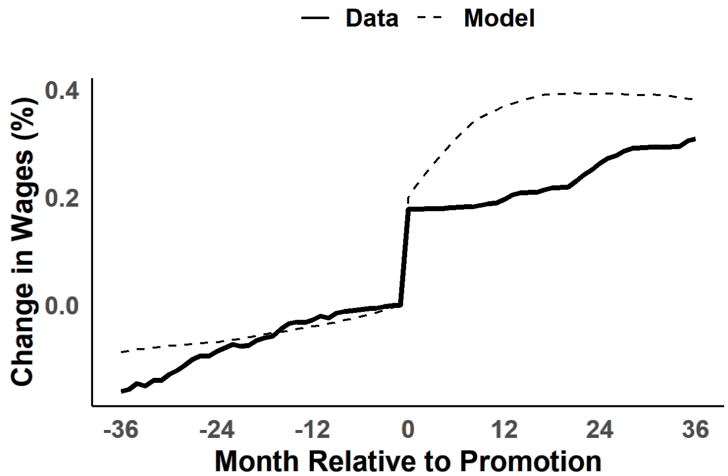


Figure 4.2: Wage Effect of Promotions: Model and Data

Notes: Figure represents log-wages around all first promotions in the NLSY and simulated data from the model. Dashed line is the model and solid line is the data - identical to that shown in Figure 2.3(b).

atically understates dispersion, producing a simulated standard deviation of 0.1637. Row two of Table 4.3 shows that the miss is concentrated in the upper tail: the model matches percentiles up to (and including) the median reasonably well, but it understates large promotion gains. Even so, this is a notable improvement over the canonical promotion model, whose implied distribution is shown in the third row of Table 4.3.

Table 4.3: Distribution of Promotions: Model and Data

	p10	p20	p50	p80	p90
Data	-12.2	1.6	14.7	34.8	51.8
Model	-9.2	0.9	16.7	27.9	32.20
Model$_{\beta^p \approx \infty}$	5.90	8.00	14.40	23.3	28.3

Notes: Values represent the 10^{th}, 20^{th}, 50^{th}, 80^{th}, and 90^{th} percentiles of promotion-induced wage gains.

The parameters α and ξ target the distribution of unemployment duration for both high and low types. In the data, average unemployment duration is 8.25 months for high types and 11.41 months for low types. In the simulated data, the corresponding averages are 7.27 and 11.14 months.

Figure 4.4 presents the model fit around unemployment-to-employment (U2E) transitions. The targeted moments are the discontinuous drops at time 0 (i.e., at re-employment), which the model matches closely. As an untargeted moment, the model does reasonably well in matching the path of wages leading into the unemployment spell, but it cannot quite match the path of wages after re-entry into employment.

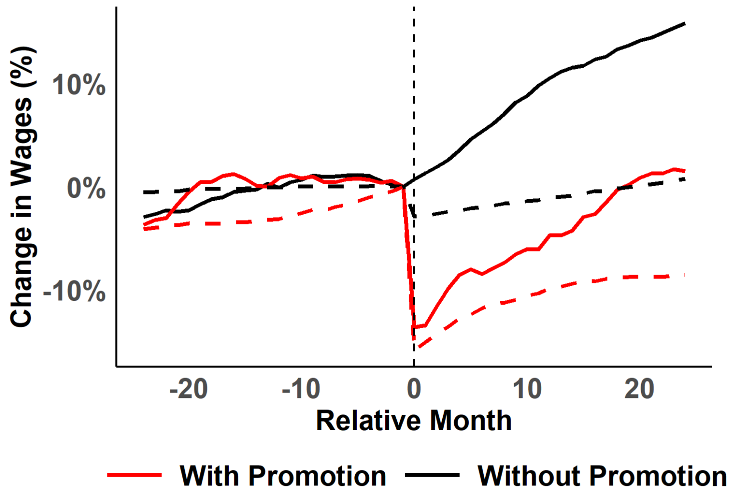
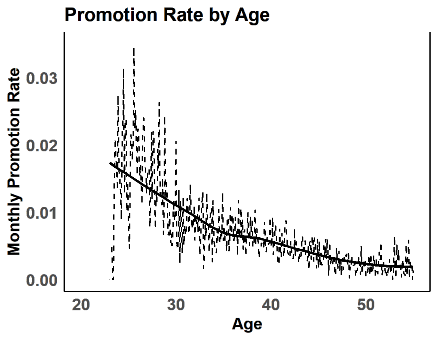


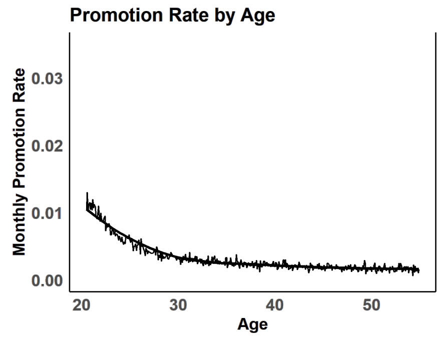
Figure 4.3: Wage Effect of Promotions: Model and Data
Notes: Figure represents log-wages around all each unemployment spell in both the model and the data. Dotted lines represent simulated data from the model, while solid lines are data pulled from the NLSY.

The parameter k_1 is estimated to match the share of workers who are promoted prior to entering unemployment—that is, the relative size of the promoted group in the U2E event studies. Currently, the model understates this promotion rate: the empirical share is about 34%, whereas the simulated share is just over 19%. However, as an untargeted moment, the model does quite well in matching the life-cycle of promotion rates. Figure 4.4 plots both the empirical (4.4(a)) and the simulated (4.4(b)) promotion rate by age. The model is capable of replicating the steep decline of promotions early in the career, though it does move to the bottom ($\sim 3\%$) earlier than the data.

The information mechanism also allows the model to naturally match the empirical interaction between promotions and unemployment duration. Figure 4.5 plots the simulated analog of Figure 2.8. While the model underpredicts the depreciation rate for both groups, it captures the key qualitative fact that unemployment duration harms previously promoted workers much more than it harms non-promoted workers.



(a) Data



(b) Model

Figure 4.4: Promotion Rate by Age: Model and Data

Notes: Figures provide the monthly rate of promotion from the data (a) and the model (b). While the model under-predicts the frequency of promotions as a whole, it can match the shape without specifically targeting it.

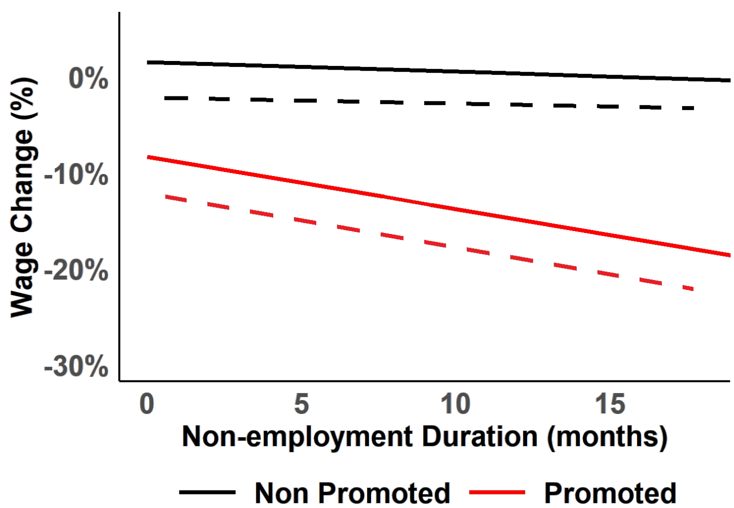


Figure 4.5: Wage Loss and Duration: Model and Data

Notes: The figure plots the relationship between unemployment duration and promotional status - both in the model (solid) and in the data (dashed).

5 Conclusion

This paper documents a set of empirical facts about promotions and their longer-run interaction with job mobility and unemployment using the NLSY79, and proposes a model of promotions and firm learning to rationalize these patterns. Consistent with the existing literature, promotions are associated with large contemporaneous wage gains and an acceleration in subsequent wage growth. At the same time, the data reveal substantial dispersion in wage changes at promotion, including meaningful wage declines for a non-trivial fraction of workers.

The central new results concern how promotion status shapes the consequences of worker churn. Promoted and non-promoted workers look broadly similar around employer-to-employer transitions that do not involve an unemployment spell, suggesting that promotion status alone is not a sufficient statistic for gains from job-to-job reallocation. In contrast, promoted workers experience large wage declines when they enter unemployment, and these losses grow rapidly with unemployment duration, whereas non-promoted workers exhibit little average wage loss and substantially weaker duration dependence. These facts are difficult to reconcile with frameworks in which the outside market observes the same information as the current employer or in which unemployment is simply a temporary interruption of tenure.

The model developed in this paper provides a simple interpretation. Promotions are informative public signals that compress a rich internal performance history into a coarse resume event. Because the current employer observes production while the outside market does not, the labor market learns about workers through observable outcomes—promotion, retention, separation, and job-finding success. In this environment, a promotion can raise wages on impact and still increase exposure to adverse inference after job loss: once unemployed, promoted workers face sharper downward revisions in market beliefs as unemployment persists, both because their prior promotion raised the market's expectations and because duration is itself informative through endogenous search. In simulations, these mechanisms generate the asymmetry between promoted and non-promoted workers after unemployment, while also matching the average wage jump and the dispersion in wage changes at promotion through stochastic promotion decisions.

Looking forward, the results suggest several directions for future work. On the empirical side, richer measurement of what firms and markets observe—for example, more detailed job-title changes, within-firm task reassignments, or performance proxies—could sharpen what information is revealed by promotion itself versus by other resume events. On the theoretical side, extending the model to include on-the-job search and multi-firm job ladders would allow promotions to interact more directly with outside offers, and could help reconcile the modest differences at E2E transitions with the stark differences after unemployment. More broadly, the paper highlights that promotions are not only a mechanism for wage growth within firms, but also a key input into how the broader labor market forms and revises beliefs about workers over their careers.

A Likelihood of Spells

$$P_N^{(\tau)}(nu \mid z, \chi) = \begin{cases} \int \Big[\delta + (1-\delta)\rho F(\pi'(y_\tau \mid \chi, p=0), 0)\Big]\, \phi(y_\tau \mid p=0, z)\, dy_\tau & \tau = nu, \\ (1-\delta) \int (1-\rho) P_N^{(\tau+1)}(nu \mid z, \chi) + & \tau \in \{1, \dots, nu-1\} \setminus \{nu\}. \\ \qquad \rho\big(1 - F(\pi'(y_\tau \mid \chi, p=0), 0)\big) \times & \\ \qquad \big(1 - \Pr(\pi'(y_\tau \mid \chi, p=0))\big) \times & \\ \qquad P_N^{(\tau+1)}(nu \mid z, \pi'(y_\tau \mid \chi, p=0))\, \phi(y_\tau \mid p=0, z)\, dy_\tau & \end{cases} \tag{A.1}$$

$$P_P^{(\tau)}(nu, np \mid z, \chi) = \begin{cases} \int \Big[\delta + (1-\delta)\rho F(\pi'(y_\tau \mid \chi, p=1), 1)\Big] & \tau = nu + np, \\ \quad \times \phi(y_\tau \mid p=1, z)\, dy_\tau & \\ (1-\delta) \int (1-\rho) P_P^{(\tau+1)}(nu, np|z, \chi) + & \tau \in \begin{matrix}\{nu+1, ..., nu+np-1\}/ \\ \{nu+np\}\end{matrix}, \\ \qquad \rho\big(1 - F(\pi'(y_\tau \mid \chi, p=1), 1)\big) \times & \\ \qquad P_P^{(\tau+1)}(nu, np \mid z, \pi'(y_\tau \mid \chi, p=1))\, \phi(y_\tau \mid p=1, z)\, dy_\tau & \\ (1-\delta)\rho \int \big(1 - F(\pi'(y_\tau \mid \chi, p=0), 0)\big) & \tau = nu, \\ \qquad \times \Pr(\pi'(y_\tau \mid \chi, p=0)) & \\ \qquad \times P_P^{(\tau+1)}(nu, np \mid z, \pi'(y_\tau \mid \chi, p=0)) & \\ \qquad \times \phi(y_\tau \mid p=0, z)\, dy_\tau & \\ (1-\delta) \int (1-\rho) P_P^{(\tau+1)}(nu, np|z, \chi) + & \tau \in \{1, ..., nu-1\}/\{nu\}. \\ \qquad \rho\big(1 - F(\pi'(y_\tau \mid \chi, p=0), 0)\big) \times & \\ \qquad \big(1 - \Pr(\pi'(y_\tau \mid \chi, p=0))\big) \times & \\ \qquad P_P^{(\tau+1)}(nu, np \mid z, \pi'(y_\tau \mid \chi, p=0))\, \phi(y_\tau \mid p=0, z)\, dy_\tau & \end{cases} \tag{A.2}$$

Equations A.1 and A.2 calculate the probability that a employed worker of type z works in a non-promoted position for nu periods, works in a promoted position (if applicable) for np periods, and then loses his job, given that he started with an initial belief of χ.

As an example, suppose a worker enters the unemployment market after accruing $nu = 2$ and $np = 2$. This worker is one that got promoted at the end of his second period, and lost his job at the end of his fourth. Let χ be the belief he entered this unemployment spell with. The value $P_P^{(0)}(2, 2|z, \chi)$ represents the likelihood (nu, np) happened to a worker like this who is type z. Expanding $P_P^{(0)}(2, 2|z, \chi)$ as defined in equation A.2 would result in a size-4 nested integral where each step in represents the event that happened at the workers k^{th} period of tenure, integrated over all paths that may have led there and weighted by their probabilities.

The recursive nature of this problem allows a programmer to build a mapping of (nu, np, z, χ) into new beliefs via backward induction.

B Existence of a Fixed Point Solution

This appendix establishes the existence of a fixed point $(V_u, V_e^P, V_e^{NP}, m^*, n^*)$ of the full system. We set $\lambda = 0$ throughout; this entails no loss of generality but suppresses q as a state variable. Write the value function compactly as

$$V(z, \pi, \chi, nu, np, j) \;=\; \begin{cases} V_u(z, \chi), & j = 1, \\ V_e^{NP}(z, \pi, \chi, nu), & j = 2, \\ V_e^{P}(z, \pi, \chi, nu, np), & j = 3, \end{cases}$$

and let $S = Z \times [0,1] \times [0,1] \times \mathbb{N}_0 \times \mathbb{N} \times \mathcal{J}$ be the state space, where $Z = \{0,1\}$ and $\mathcal{J} = \{1,2,3\}$ carry the discrete topology and $[0,1]$ the standard topology. As a finite product of compact spaces, S is compact, so $C(S)$—the space of continuous real-valued functions on S with sup-norm $\|f\| = \sup_{s\in S} |f(s)|$—is a Banach space.

The argument proceeds via three lemmas. Lemma B.4 establishes that the Bellman operator is a contraction for fixed belief transition functions, delivering a unique value function $V^*(m,n)$. Lemma B.5 shows that $V^*(m,n)$ and the induced optimal policy $\hat{\theta}^*(m,n)$ vary continuously with (m,n). Lemma B.6 constructs a compact convex set K_{L^*} that is mapped into itself by the belief-updating operator Ψ. Theorem B.8 then applies Schauder's fixed point theorem.

We maintain the following assumptions.

Assumption B.1. The functions $p(\theta, z)$, $\nu(\theta)$, $w(\pi, p)$, and $\Pr(\pi)$ are continuous in their arguments. Moreover, $\nu(\theta)$ is strictly convex and twice continuously differentiable, with $\nu(\theta) \to \infty$ as $\theta \to \infty$.

Assumption B.2. The matching probability satisfies

$$p(\theta, z) \;=\; \underline{p} + (1-\underline{p})\big(1 - e^{-\alpha(z)\theta}\big) \;\in\; (\underline{p},\, 1),$$

where $\underline{p} > 0$ and $\alpha(z) \geq 1$ for all $z \in Z$. Set $\alpha^{\max} = \max_{z\in Z} \alpha(z)$ and $L_p^{\max} = (1-\underline{p})\alpha^{\max}$; the latter serves as a uniform Lipschitz constant for p in θ.

Assumption B.3. The optimal search effort satisfies $\hat{\theta}^*(z, \chi; m, n) > 0$ for all (z, χ, m, n).

Assumption B.3 is implied by Assumption B.2: at $\theta = 0$ the marginal benefit of search equals $\alpha(z)\beta(1-d)[\cdots] > 0$ while $\nu'(0)$ is finite, so the optimizer always chooses strictly positive effort.

Lemma B.4 (Existence and uniqueness of V^*)**.** *For any fixed $(m,n) \in C([0,1],[0,1])^2$, the Bellman operator $T : C(S) \to C(S)$ is a contraction with modulus $\tilde{\beta} \equiv \beta(1-d) \in (0,1)$. Consequently, there exists a unique $V^*(m,n) \in C(S)$ satisfying $TV^*(m,n) = V^*(m,n)$.*

Proof. We verify Blackwell's sufficient conditions.

Monotonicity. Fix $v, t \in C(S)$ with $v \leq t$ pointwise, and let v_j, t_j denote their restrictions to employment state j. For $j = 1$, let $\hat{\theta}$ attain the maximum in (Tv):

$$
\begin{aligned}
(Tv)(z, \pi, \chi, nu, np, 1) &= b - \nu(\hat{\theta}) + \tilde{\beta}\big[p(\hat{\theta}, z)\Pr(m(\chi))\, v_3(z, m(\chi), m(\chi), 0, 0) \\
&\qquad + p(\hat{\theta}, z)(1 - \Pr(m(\chi)))\, v_2(z, m(\chi), m(\chi), 0) \\
&\qquad + (1 - p(\hat{\theta}, z))\, v_1(z, n(\chi))\big] \\
&\leq b - \nu(\hat{\theta}) + \tilde{\beta}\big[p(\hat{\theta}, z)\Pr(m(\chi))\, t_3(z, m(\chi), m(\chi), 0, 0) \\
&\qquad + p(\hat{\theta}, z)(1 - \Pr(m(\chi)))\, t_2(z, m(\chi), m(\chi), 0) \\
&\qquad + (1 - p(\hat{\theta}, z))\, t_1(z, n(\chi))\big] \\
&\leq \max_{\theta}\Big\{b - \nu(\theta) + \tilde{\beta}\big[p(\theta, z)\Pr(m(\chi))\, t_3(z, m(\chi), m(\chi), 0, 0) \\
&\qquad\qquad + p(\theta, z)(1 - \Pr(m(\chi)))\, t_2(z, m(\chi), m(\chi), 0) \\
&\qquad\qquad + (1 - p(\theta, z))\, t_1(z, n(\chi))\big]\Big\} \\
&= (Tt)(z, \pi, \chi, nu, np, 1).
\end{aligned}
$$

The first inequality uses $p(\cdot), \Pr(\cdot) \geq 0$ and $v \leq t$ pointwise; the second uses feasibility of $\hat{\theta}$ for t. The argument is identical for $j = 2, 3$, establishing $Tv \leq Tt$.

Discounting. For any $v \in C(S)$ and constant $a \geq 0$, write $\mathbf{a}$ for the function identically equal to a:

$$
\begin{aligned}
T(v + \mathbf{a})(s) &= \max_{\theta}\big[b - \nu(\theta) + \tilde{\beta}\,\mathbb{E}[(v + \mathbf{a})(s') \mid \theta, s]\big] \\
&= \max_{\theta}\big[b - \nu(\theta) + \tilde{\beta}\,\mathbb{E}[v(s') \mid \theta, s] + \tilde{\beta} a\big] \;=\; (Tv)(s) + \tilde{\beta} a.
\end{aligned}
$$

The same calculation holds for $j = 2, 3$. Since $\tilde{\beta} \in (0, 1)$, the discounting condition is satisfied with modulus $\tilde{\beta}$.

By Blackwell's theorem, T is a contraction on $(C(S), \|\cdot\|)$ with modulus $\tilde{\beta}$. That T maps $C(S)$ into itself follows from continuity of all primitives and Berge's maximum theorem. The Banach fixed point theorem yields the unique fixed point $V^*(m, n)$. □

Lemma B.5 (Continuity of V^* and $\hat{\theta}^*$)**.** *Let $(m_k, n_k) \to (m, n)$ uniformly in $C([0, 1], [0, 1])^2$. Then $\|V^*(m_k, n_k) - V^*(m, n)\| \to 0$, and $\hat{\theta}^*(\,\cdot\,; m_k, n_k) \to \hat{\theta}^*(\,\cdot\,; m, n)$ uniformly.*

Proof. Continuity of V^*. Since $V^*(m_k, n_k) = T(V^*(m_k, n_k); m_k, n_k)$ and $V^*(m, n) =$

$T(V^*(m,n);m,n)$, the triangle inequality gives

$$\|V^*(m_k,n_k) - V^*(m,n)\| \le \underbrace{\|T(V^*(m_k,n_k);m_k,n_k) - T(V^*(m_k,n_k);m,n)\|}_{\text{(I)}}$$
$$+ \underbrace{\|T(V^*(m_k,n_k);m,n) - T(V^*(m,n);m,n)\|}_{\text{(II)}}.$$

Applying the contraction property to term (II) and rearranging, $(1-\tilde{\beta})\|V^*(m_k,n_k) - V^*(m,n)\| \le \text{(I)}$. Term (I) measures the direct sensitivity of T to the belief functions at a fixed value function argument; the belief functions enter only through the $j=1$ equation via $m(\chi)$ and $n(\chi)$. Since $(m_k,n_k) \to (m,n)$ uniformly and $\{V^*(m_k,n_k)\}$ is uniformly bounded (the contraction bound gives $\|V^*\| \le (1-\tilde{\beta})^{-1}\|TV_0 - V_0\|$ independently of (m,n)), continuity of all primitives implies $\text{(I)} \to 0$, so $\|V^*(m_k,n_k) - V^*(m,n)\| \le (1-\tilde{\beta})^{-1}\text{(I)} \to 0$.

Continuity of $\hat{\theta}^*$. The search problem for $j=1$ is $\hat{\theta}^*(\chi,z;m,n) = \arg\max_{\theta\in[0,\Theta^{\max}]} W(\theta,\chi,z,m,n)$, where the truncation to $\Theta^{\max} < \infty$ is without loss because $\nu(\theta) \to \infty$ and V^* is bounded. The objective W is jointly continuous in (θ,χ,m,n) by continuity of p, ν, Pr, and V^*. The constraint correspondence $\Phi(\cdot) = [0,\Theta^{\max}]$ is constant, hence trivially continuous and compact-valued. By Berge's maximum theorem, $\hat{\theta}^*$ is upper hemicontinuous in (m,n). Strict concavity of W in θ—which follows from the structure of p in Assumption B.2 and strict convexity of ν in Assumption B.1—implies the maximizer is unique at every parameter value. A single-valued upper hemicontinuous correspondence is a continuous function. □

Define the belief-updating operator $\Psi : C([0,1],[0,1])^2 \to C([0,1],[0,1])^2$ by $\Psi(m,n) = (m',n')$, where

$$m'(\chi) = \frac{\chi\, a(\chi)}{\chi\, a(\chi) + (1-\chi)\, b(\chi)}, \qquad n'(\chi) = \frac{\chi(1-a(\chi))}{\chi(1-a(\chi)) + (1-\chi)(1-b(\chi))}, \tag{B.1}$$

$a(\chi) = p(\hat{\theta}^*(\chi,z{=}1;m,n),z{=}1)$, and $b(\chi) = p(\hat{\theta}^*(\chi,z{=}0;m,n),z{=}0)$. A fixed point of Ψ is a pair of belief transition functions consistent with optimal search behavior via Bayes' rule. For $L>0$, define

$$K_L = \big\{(m,n) \in C([0,1],[0,1])^2 : |m(\chi)-m(\chi')| \le L|\chi-\chi'|,\ |n(\chi)-n(\chi')| \le L|\chi-\chi'|\ \forall\, \chi,\chi'\big\}.$$

The Lipschitz condition is preserved under convex combinations, and K_L is uniformly bounded and equicontinuous, so K_L is a compact convex subset of $C([0,1])^2$ by the Arzelà–Ascoli theorem.

Lemma B.6 (Invariance)**.** *Suppose*

$$3C_2 < \underline{p}^2, \tag{B.2}$$

where

$$C_2 = \frac{(L_p^{\max})^2\beta(1-d)}{\kappa}\left(2\|V^*\|\|\,\mathrm{Pr}'\,\|_\infty + \|\,\mathrm{Pr}\,\|_\infty\left\|\frac{\partial V_3^*}{\partial m}\right\|_\infty + \|1-\mathrm{Pr}\,\|_\infty\left\|\frac{\partial V_2^*}{\partial m}\right\|_\infty + \left\|\frac{\partial V_1^*}{\partial n}\right\|_\infty\right), \tag{B.3}$$

and $\kappa > 0$ *is a lower bound on* $|\partial^2 W/\partial\theta^2|$ *at the optimum. Then* $\Psi : K_{L^*} \to K_{L^*}$ *for* $L^* = 2\underline{p}^{-2}/(1-3C_2\underline{p}^{-2}) < \infty$*.*

Proof. We bound $|dm'/d\chi|$; the argument for n' is symmetric. Write $f(\chi) = \chi a(\chi)$ and $g(\chi) = \chi a(\chi) + (1-\chi)b(\chi)$, so $m' = f/g$ and, by the quotient rule, $dm'/d\chi = [f'g - fg']/g^2$.
Denominator. Since $p \geq \underline{p}$, we have $g \geq \underline{p}$ and $g^2 \geq \underline{p}^2$.

Lipschitz bounds on a and b. The chain rule and $|\partial p/\partial\theta| \leq L_p^{\max}$ give $|a'(\chi)| \leq L_p^{\max}|d\hat{\theta}^*/d\chi|_{z=1}$ and similarly for $|b'(\chi)|$. To bound $|d\hat{\theta}^*/d\chi|$, apply the implicit function theorem to the first-order condition

$$F(\theta,\chi) \;:=\; -\nu'(\theta) + \frac{\partial p}{\partial\theta}\,\tilde{\beta}\left[\mathrm{Pr}(m(\chi))V_3^*(m(\chi)) + (1-\mathrm{Pr}(m(\chi)))V_2^*(m(\chi)) - V_1^*(n(\chi),z)\right] = 0.$$

Strict concavity of W ensures $|\partial F/\partial\theta| \geq \kappa > 0$, so the theorem applies. Differentiating F with respect to χ, using $(m,n) \in K_L$ so that $|m'|,|n'| \leq L$, and bounding all primitive terms gives $|\partial F/\partial\chi| \leq C_1 L$, where C_1 is the bracket in (B.3) multiplied by $L_p^{\max}\tilde{\beta}$. Hence $|d\hat{\theta}^*/d\chi| \leq C_1 L/\kappa$, and therefore $|a'|,|b'| \leq C_2 L$.
Collecting terms. Since $|f|,|g| \leq 1$ and $a,b \in [\underline{p},1]$:

$$|f'| \leq 1 + C_2 L, \quad |g'| \leq 1 + 2C_2 L, \quad |f'g - fg'| \leq 2 + 3C_2 L.$$

Combining with the denominator bound, $|dm'/d\chi| \leq (2+3C_2L)/\underline{p}^2$. Setting this $\leq L$ requires $L(1-3C_2/\underline{p}^2) \geq 2/\underline{p}^2$, which has a finite positive solution if and only if (B.2) holds. The choice $L^* = 2\underline{p}^{-2}/(1-3C_2\underline{p}^{-2})$ then satisfies $|dm'/d\chi| \leq L^*$, and the symmetric bound for n' is identical, so $\Psi(K_{L^*}) \subseteq K_{L^*}$. □

Remark B.7. Condition (B.2) jointly restricts $\underline{p}$, $\alpha^{\max}$, β, d, and κ. Because C_2 depends on value function derivatives determined at the fixed point, the condition cannot be verified analytically, but it can be checked numerically. Economically, it requires the baseline matching probability $\underline{p}$ to be sufficiently large relative to the sensitivity of optimal search effort to changes in beliefs.

Theorem B.8 (Existence)**.** *Under Assumptions B.1–B.3 and condition* (B.2)*, there exists a fixed point* $(m^*,n^*) \in K_{L^*}$ *of* Ψ *and a corresponding solution* $(V_u, V_e^P, V_e^{NP}, m^*, n^*)$ *to the full system in which the value functions solve the Bellman equations and the belief transition functions are consistent with optimal search behavior via Bayes' rule.*

Proof. K_{L^*} is compact and convex by construction, and $\Psi(K_{L^*}) \subseteq K_{L^*}$ by Lemma B.6. Continuity of Ψ on K_{L^*} follows from Lemma B.5 (for $(m,n) \mapsto \hat{\theta}^*$), continuity of $p(\cdot, z)$ (for $\hat{\theta}^* \mapsto (a,b)$), and smoothness of the Bayesian formulas in (B.1) with denominators bounded below by $\underline{p} > 0$ (for $(a,b) \mapsto (m', n')$). By Schauder's fixed point theorem, Ψ has a fixed point $(m^*, n^*) \in K_{L^*}$. Lemma B.4 then yields the unique $V^*(m^*, n^*) \in C(S)$ solving the Bellman equations at (m^*, n^*), completing the proof. $\square$

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